

High Density Modular Multilevel Cascade Converter for Medium-Voltage Motor Drive

Dong Jiang, Jing Xue, Fred Wang

Min H. Kao Department of Electrical Engineering and Computer Science
The University of Tennessee, Knoxville
Knoxville, TN, 37914
djiang4@utk.edu

Abstract—This paper proposes a new Modular Multilevel Cascade Converter (MMCC) topology for future high density medium-voltage motor drive. The proposed topology utilizes the active ripple energy storage method for rectifier capacitance design. Simulation has been done in this paper and proves that the proposed topology works well, and the active energy storage method can significantly reduce the DC-link capacitor. Power density improvement is quantified for an example system to show the benefits of the proposed topology.

I. INTRODUCTION

Modular Multilevel Cascade Converter (MMCC) has been proposed as solution of high density motor drive in railway traction and electric ship propulsion application since early 2000's [1]. The basic idea of MMCC is building the motor drive based on independent converter modules. High voltage and high power can be achieved with low voltage power electronics devices in each of the power modules. Another advantage of MMCC is that high frequency transformers are used to replace the conventional line-frequency transformer[3][4], which can significantly reduce the weight and volume of the system.

Despite the volume and weight benefits, MMCC still suffers bottleneck for further power density improvement from passives like DC capacitors and magnetics in solid-state transformer. In previous work [5],[6], active ways to reduce the energy storage capacitor of rectifier have been studied. This paper utilizes this concept in MMCC design for possible higher power density.

II. TOPOLOGY ANALYSIS

Fig.1 shows the topology of the recent proposed modular multilevel cascade converter as motor drive^{[2],[3]}. In each of the cells, the input side is a single-phase rectifier, then a dual-active-bridge (DAB) module transfers the power to the output side and also plays the role of isolation, the output side is a single-phase inverter generating PWM voltage waveform for the load. By connecting N cells of the modules, $2N+1$ level of output voltage can be achieved for the motor drive phase-voltage. The DAB switches with much higher frequency than line-frequency and the size of the single-phase transformer can be much smaller than grid side multi-pulse isolation transformer used in conventional cascade multilevel motor drive.

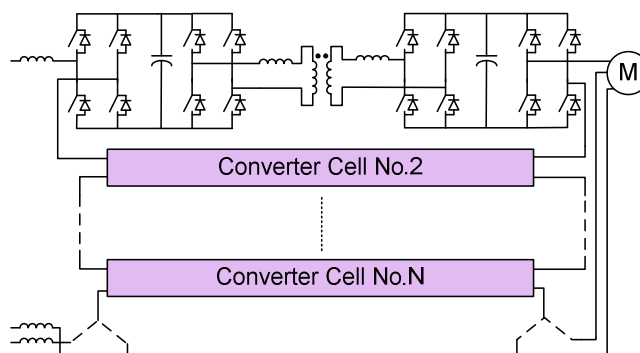


Figure 1. Bidirectional DC-DC converter based medium-voltage motor drive proposed in reference [3]

The principle of the DAB is shown in Fig.2. The gate pulses of the primary and secondary side H-bridge have the same duty cycle but with a phase-shift angle. The transformer current follows the curve of Fig.2 and transfers power to the secondary side.

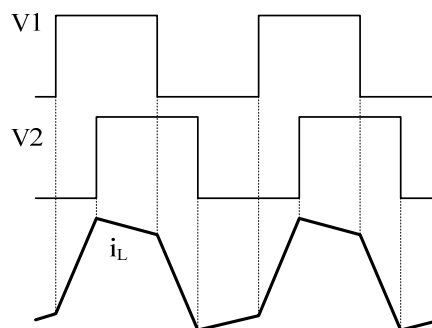


Figure 2. Voltage and transformer current waveform in the bidirectional DC-DC converter

Because of the second-order harmonics embedded in the single-phase rectifier, there is an obvious low frequency harmonics in the DC-link voltage and requires big DC capacitor to absorb it, which is a big contributor to the overall system volume and weight. Reference [5] and [6] proposed the single-phase rectifier with active energy storage, shown in Fig.3. An auxiliary phase-leg is connected to L_s and C_s , by switching the auxiliary phase-leg, the DC voltage ripple energy could be absorbed by L_s - C_s . Control of the auxiliary circuit is shown in Fig.4, the error of the DC-link voltage is controlled by two comparator and generated gate signals for the top switch G5 and bottom switch G6.

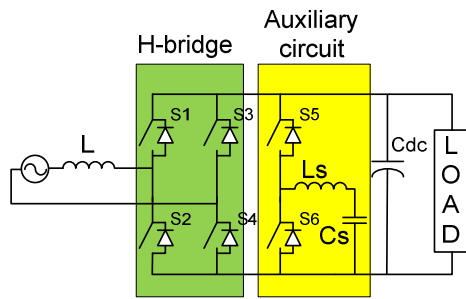


Figure 3. Single-phase rectifier with active energy storage

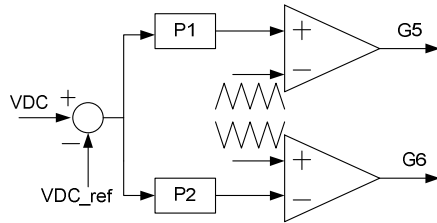


Figure 4. Voltage ripple control method for the auxiliary bridge

Based on the concept of active energy storage, Fig.5 shows the improved MMCC converter based medium-voltage motor drive. Different from Fig.1, in each cell, the input and output side single-phase converter is assisted by the active energy storage phase-leg. With this topology, the DC-link capacitor can be significantly reduced, leading to increased power density.

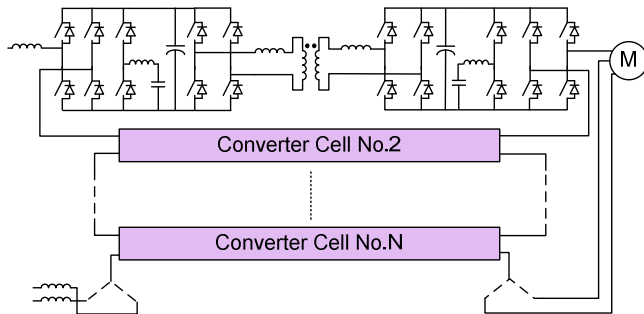


Figure 5. Bidirectional DC-DC converter based medium-voltage motor drive with active energy storage

III. SIMULATION ANALYSIS

Simulation in MATLAB/Simulink has been done for an example system to evaluate the feasibility and benefit of the proposed multi-level cascade medium-voltage topology. Single-stage is used for simulation as shown in Fig.6. The parameters are shown in TABLE I. The input side is a single-phase voltage source with 380V RMS voltage and controlled with unity power factor. The load side inverter is connected to a L-R load with open-loop control. The DC-link voltage of the load side is 720V. 1200V/100A IGBTs are used as the switches and the converter could serve as one cell in a five-level 4160V motor drive. In simulation, 10 kHz switching frequency is applied to the switches.

In order to show the benefits of the active energy storage, after 0.9s, the gate signals of the auxiliary phase-leg are blocked both for input and output side.

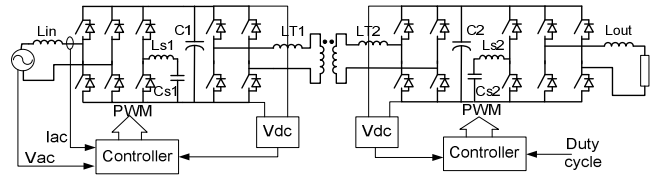


Figure 6. Simulation system: single block of the AC-AC power converter with dual-active-bridge

TABLE I. SYSTEM PARAMETERS

Input voltage	380V (RMS)
DAB primary voltage	650V
DAB secondary voltage	720V
DC-link capacitor (C1, C2)	0.4mF
Auxiliary inductor (Ls)	40μH
Auxiliary capacitor (Cs)	0.23mF
Transform leakage inductance (LT1+LT2)	200μH

Simulation results are shown in Fig.7-Fig.11. Fig.7 shows the waveforms in the input side, before 0.9s, G5 and G6 are switching and charging the capacitor Cs with 150-650V voltage, the DC-link voltage is controlled with ripple peak-to-peak value less than 30V. After 0.9s, the gate signals are blocked and the auxiliary capacitor is stopped being charged, the second-order ripple obviously increased in the DC-link voltage, which is higher than 200V. Similar phenomenon can be found in the load side converter, shown in Fig.8.

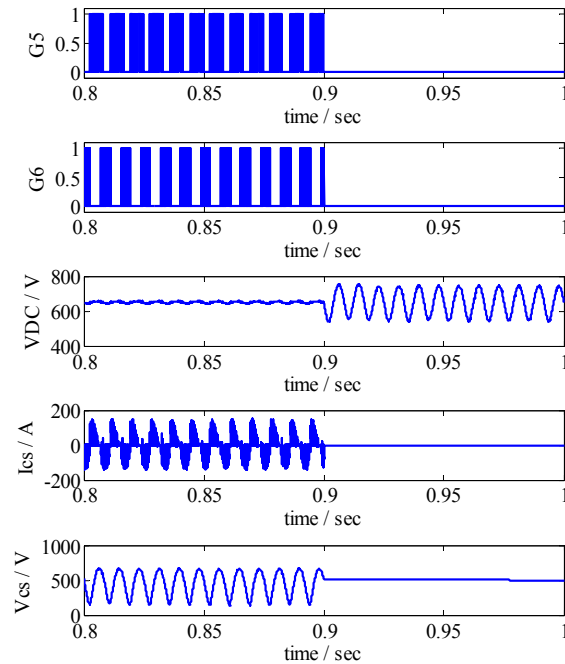


Figure 7. Simulation results: source side converter

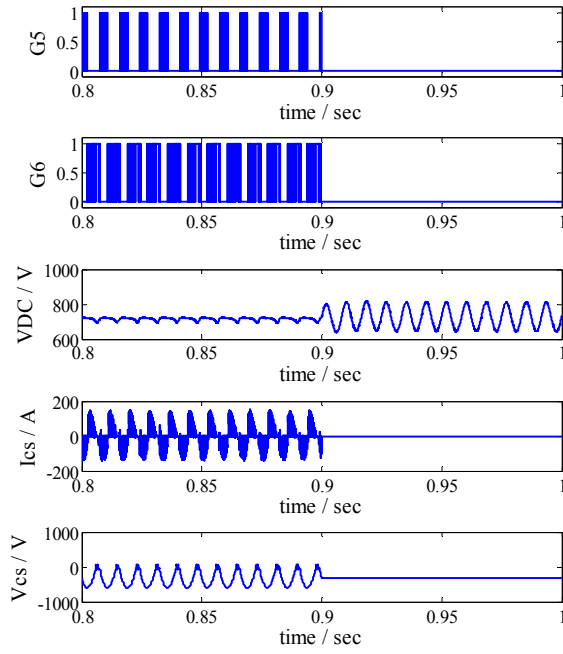


Figure 8. Simulation results: load side converter

The output voltage is shown in Fig.9, by switching the output single-phase inverter, a typical three-level voltage is generated and could contribute to the multilevel inverter. Without active energy storage (after 0.9s), the output DC voltage has bigger ripple and also influences the output voltage.

Fig.10 shows the DAB primary and secondary side voltage and the current in the transformer. The simulation results fit the principle in Fig.2 well.

Fig.11 shows the input voltage and input current, unity power factor is achieved in the converter.

Fig.12 shows the output current with R-L load (1mH, 5Ω), 18kW active power is achieved by this cell.

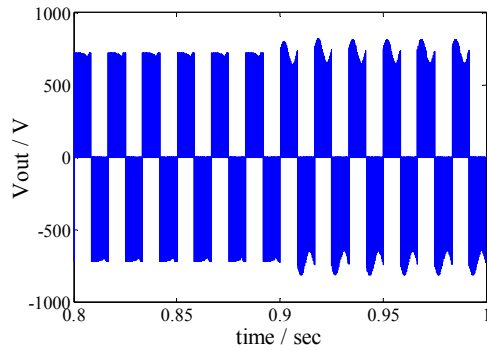


Figure 9. Simulation result: output voltage

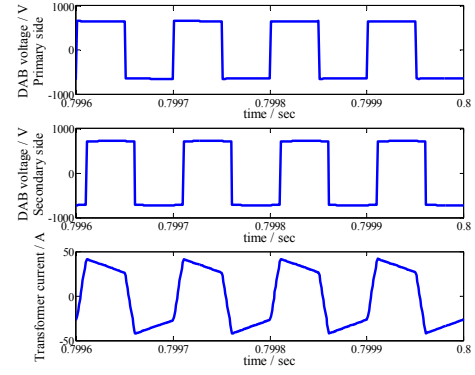


Figure 10. Simulation results: DAB voltage and current

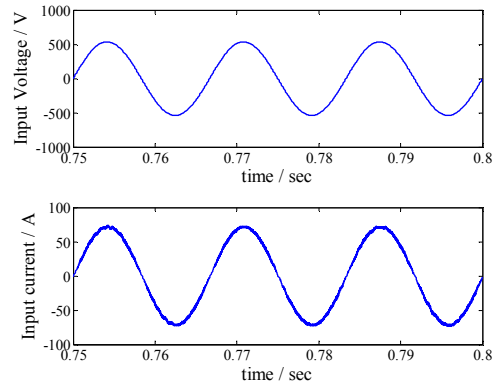


Figure 11. Simulation results: input voltage and current

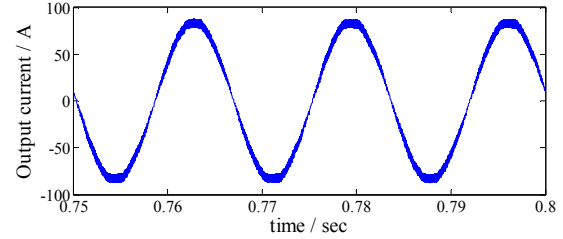


Figure 12. Output current with RL load (1mH, 5Ω)

IV. WEIGHT COMPARISON

By the simulation results, for 30V peak-to-peak ripple requirement in the DC-link voltage, without active energy storage, DC-link capacitor should be no less than 2.3mF. With auxiliary circuit, 0.4mF DC capacitor and auxiliary components of $L_s=40\mu\text{H}$ and $C_s=0.23\text{mF}$.

As a preliminary design, two of the 230 μF film capacitors 478-5981-ND are selected as the auxiliary capacitor for both the input and output side.

And two of 478-5981-ND capacitors are installed in parallel on each side as DC bus capacitor for the system with auxiliary circuit. As a comparison, for the 2.3 mF DC bus capacitor in circuit without active energy storage, 10 of the 478-5981-ND are selected for each side.

For the auxiliary inductor, two Ferrite toroid cores 0P44932TC are selected for the two sides of the system, from MAGNETICS®. And for each inductor, 3 turns of AWG#8 wires are wound.

Because of the active energy storage technology, two more phase legs with their heatsink have to be installed in the system. According to [5], the heatsink of the switches are designed and 2,000 gram of weight saving by active energy storage has been estimated.

With the assumption of same DC bus voltage conditions, the isolation transformers for both systems with and without active energy storage share the same design. Two Ferrite U shape cores are selected as 0P49925UC and AW#8 wires are wound for 64 turns for both primary and secondary side.

The summarized weight comparison of the power cell with and without active energy storage auxiliary circuit is shown in the following TABLE II:

TABLE II. PASSIVE WEIGHT COMPARISON

Weight (g)	w/o AES	w/ AES
DC Bus Cap	14,000	2,800
DAB Transformer	2,220	2,220
Aux. Inductor	N/A	340
Aux. Capacitor	N/A	1,400
Extra Heatsink	5,120	3,120
Total Weight	16,220	8,760

From TABLE II, it is found that with the utilization of active energy storage technology, the passive weight of the power cell can be reduced up to 38.6%. The DC bus capacitor has the biggest weight saving of 80% as shown in Fig. 13.

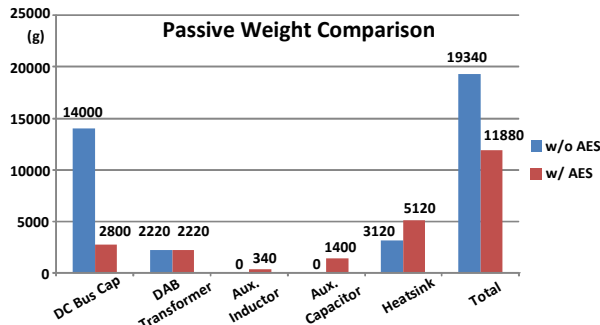


Figure 13. Passive Weight Comparison

The passive weight distribution of the two systems can be found in Fig. 14:

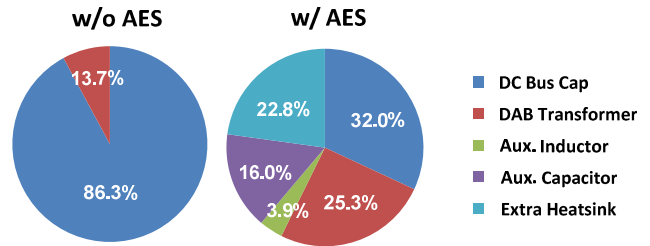


Figure14. Weight distribution for the system

V. CONCLUSIONS

A new Modular Multilevel Cascade Converter (MMCC) topology for future high density medium-voltage motor drive has been proposed in this paper. With the utilization of active ripple energy storage method for rectifier and inverter capacitance design, the proposed topology can largely increase the system power density by reducing the weight of passive components. Systematic simulation results prove that the proposed topology can achieve much smaller DC-link voltage ripple and generate stable three-level output voltage in each of the converter cell. Preliminary design of the passive components has been done in this paper and proves that up to 38.6% of the passive weight can be saved totally in the proposed application.

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REFERENCE

- [1] M. Hagiwara, H. Akagi, "Control and Experiment of Pulsewidth-Modulated Modular Multilevel Converters," *IEEE Trans. on Power Electronics*, vol.24, no.7, pp.1737-1746, July 2009.
- [2] R. W. De Doncker, D. M. Divan, M. H. Kheraluwala, "A three-phase soft-switched high-power-density DC/DC converter for high-power applications," *IEEE Trans. Ind. Applicat.*, vol.27, no.1, pp.63-73, Jan/Feb 1991.
- [3] S. Inoue, H. Akagi, "A Bidirectional Isolated DC-DC Converter as a Core Circuit of the Next-Generation Medium-Voltage Power Conversion System," *IEEE Transactions on Power Electronics*, vol.22, no.2, pp.535-542, March 2007.
- [4] S. Inoue, H. Akagi, "Voltage Control of a Bi-Directional Isolated DC/DC Converter for Medium-Voltage Motor Drives," in *Proc. Power Conversion Conference*, pp.1244-1250, 2-5 April 2007.
- [5] R. Wang, F. Wang, D. Boroyevich, P. Ning, "A high power density single phase PWM rectifier with active ripple energy storage," in *Proc. IEEE APEC'10*, pp.1378-1383, 21-25 Feb. 2010.
- [6] R. Wang, F. Wang, R. Lai, P. Ning, R. Burgos, D. Boroyevich, "Study of Energy Storage Capacitor Reduction for Single Phase PWM Rectifier," in *Proc. IEEE APEC'09*, pp.1177-1183, 15-19 Feb. 2009.